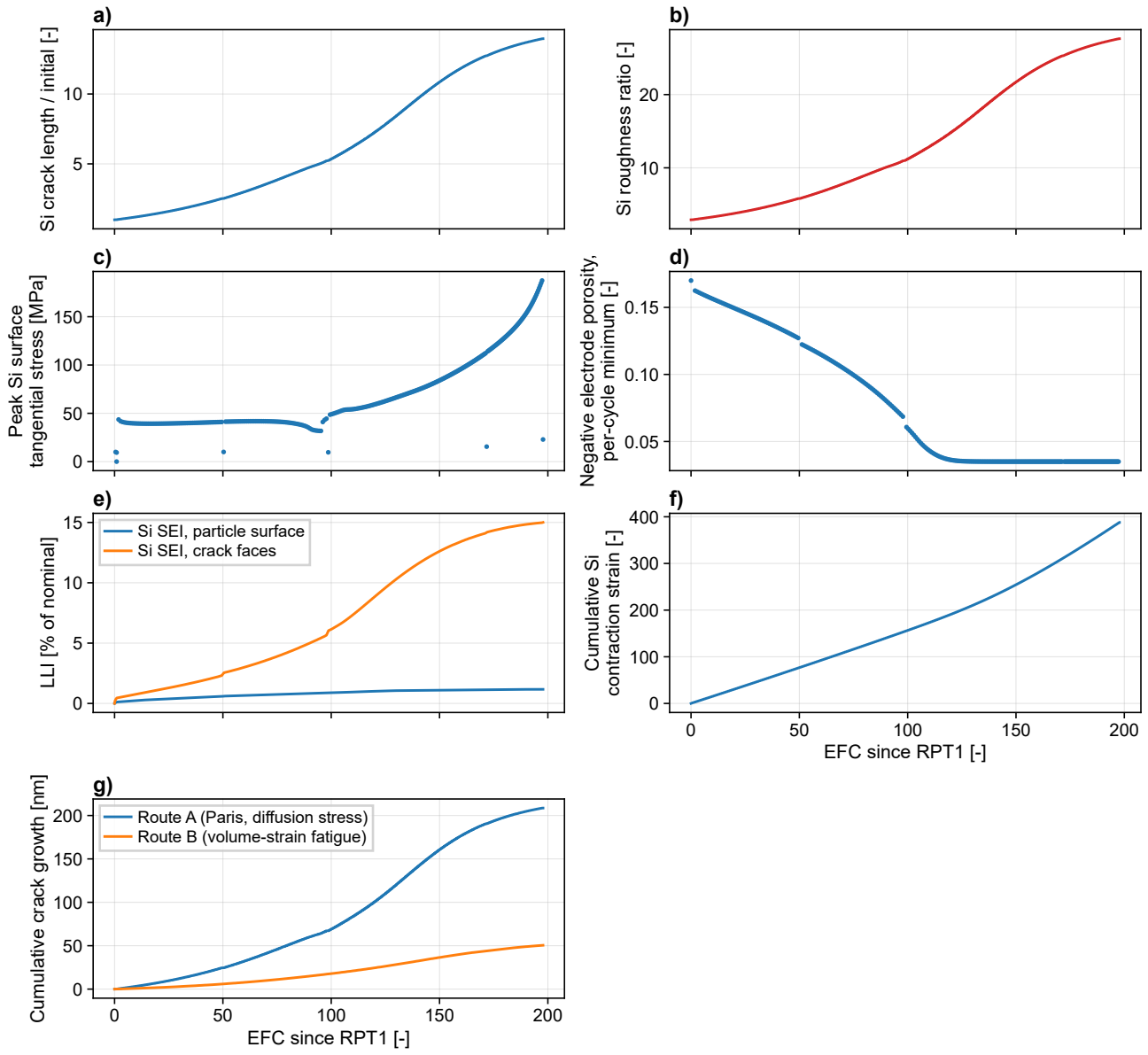

a)
Si crack length / initial [-]
10
5
b)
Si roughness ratio [-]
20
10
c)
Peak Si surface tangential stress [MPa]
150
100
50
0
d)
Negative electrode porosity, per-cycle minimum [-]
0.15
0.10
0.05
e)
LLI [% of nominal]
15
10
5
0
Si SEI, particle surface
Si SEI, crack faces
f)
Cumulative Si contraction strain [-]
400
300
200
100
0
0
50
100
150
200
EFC since RPT1 [-]
g)
Cumulative crack growth [nm]
200
150
100
50
0
Route A (Paris, diffusion stress)
Route B (volume-strain fatigue)
0
50
100
150
200
EFC since RPT1 [-]